

Uday Kiran Padhy

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Summary

Senior Software Engineer at Gainsight working at the intersection of LLM-powered product features and high-throughput backend infrastructure -- Python and FastAPI services, AI multi-agent orchestration, distributed systems on AWS and GCP.

Experience

Gainsight — Hyderabad, India — August 2022 -- Present

Promoted twice: Associate Software Engineer → Software Engineer → Senior Software Engineer.

Senior Software Engineer — Aug 2025 -- Present

- Shipped 5+ enterprise grade features that landed Fortune 500 accounts including Intuit and Adobe, sustained 99.95% measured uptime across high stakes deployments.
- Promoted internal APIs to public facing endpoints under proper authorization, enabling customers to integrate the platform into their own systems.
- Built admin dashboards over 5M+ data points with multi criteria filtering, writes propagate through an event driven pipeline keeping reads consistent within a fraction of a second.
- Implemented 3 production agents (Moderation, Search, Translation) with stateful multi-step reasoning, tool use, and conditional graph routing -- 95% decision accuracy on previously manual workflows; CloudWatch observability sustained 99.9% uptime.

Software Engineer — Aug 2023 -- Aug 2025

- Designed high-throughput APIs serving 2M+ requests/day at ~500 RPS peak with P99 latency under 200ms; drove cost per request down 30% via caching, architecture iteration, and query plan tuning.
- Rolled out SSO across the multi-product platform with 5 identity providers (SAML 2, OAuth2, OIDC, Token, Google); replaced per-product credentials with a unified sign-in surface.
- Extracted 2 team-owned services from the monolith into independently deployable FastAPI microservices with dedicated scaling and CI; cut deployment time from ~30 min to ~10 min and enabled per-service hotfixes.

Associate Software Engineer — Aug 2022 -- Aug 2023

- Built a rule based user segmentation engine with multi attribute predicates (registration recency, role, post counts), powers targeted audiences across the platform.
- Automated CloudFormation IaC for 2 microservices (~83 AWS resources spanning Lambda, RDS, IAM) with PR driven deploys.
- Wrote 200+ unit, integration, and E2E tests for critical services; increased coverage from ~40% to ~75%, catching regressions pre-merge.

Projects

Aakar — LLM Architecture Visualizer — FastAPI, Python, React, TypeScript, Redis, Docker, GCP Cloud Run

- Full-stack LLM-architecture visualizer live at aakar.udaykp.dev: FastAPI backend introspects any HuggingFace model without downloading weights; React + React Flow frontend renders the module tree as an interactive, zoomable diagram; layered clean architecture with Strategy/Registry extension points, disk + Redis two-tier cache, multi-stage Docker images deployed on Cloud Run.

AWS RL Environment — Python, GRPO, SFT, Unsloth, FastAPI, OpenEnv, HuggingFace

- Built a production-grade RL training pipeline for cloud agents -- 120+ YAML-defined AWS tasks across 5 difficulty tiers, 8-way parallel rollouts on a single consumer GPU, full SFT → GRPO pipeline backed by a vendored MiniStack simulator (34 AWS services, zero cloud cost); lifted Qwen2.5-Coder-3B from 39% to 89% exact-match accuracy after SFT (+6 pp from GRPO), and published HuggingFace adapters, dataset, and a live Space.

Technical Skills

Languages: Python, TypeScript

Backend & Distributed Systems: FastAPI, Microservices, Event-Driven Architecture, Server-Sent Events, Authentication & Authorization, System Design

Cloud & Infrastructure: AWS, GCP, Docker, CloudFormation, Infrastructure as Code, Observability & Monitoring

Data & AI Foundations: PostgreSQL, NoSQL, ETL & Data Modeling, Vector Retrieval, PyTorch, HuggingFace, SFT & GRPO

Education

Sagi Rama Krishnam Raju Engineering College — Bhimavaram, Andhra Pradesh — June 2018 -- July 2022

Bachelor of Technology in Electronics and Communication

Achievements

LeetCode Weekly Contest 393: Rank 233 worldwide; profile leetcode.com/u/IamUday.

Smart India Tech Hackathon: 6th place; led a team of 4 to ship an accessibility app for physically challenged users over a 24-hour build, beating 100+ teams.